KrishiGrid Bangladesh
Data-Driven Food Security Simulation

KRISHIGRID BANGLADESH

A Data-Driven Simulation of Food Import Dependency and Decentralized Agricultural Infrastructure

PART 8 — THE COMPLETE JOURNEY

I spent weeks analyzing Bangladesh's food security crisis from every angle imports, land, crops, infrastructure, economics, pilots, and global context.

Here is everything I found.

This project brings together the problem, the opportunity,
the solution, and the roadmap.

Total: 11 datasets, 788 rows, 42 queries, 7 presentations
Tools: SQL, Power BI

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Project Repository:
https://github.com/rafijurrahman/krishigrid-bangladesh
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RECAP

PART 1 — THE IMPORT CRISIS
Dataset: imports.csv | 140 rows | 6 queries

Key findings:
Wheat and edible oil dominate Bangladesh's import costs
Import dependency is concentrated in just a few products
2022 price surge driven by global shocks and BDT depreciation
25% reduction in high-risk imports could save $1B+ annually
Edible oil has no domestic production base — highest structural risk

PART 2 — THE HIDDEN LAND OPPORTUNITY
Dataset: lands.csv | 64 rows | 8 queries

Key findings:
Rice-fallow land is the largest untapped category at 802K hectares
Mymensingh, Noakhali, and Bhola lead in unused land
Cropping intensity is already high nationally
Future gains come from optimizing what is grown, not adding cycles
Targeted regional intervention delivers far greater returns than uniform national policies

PART 3 — THE CROP REVOLUTION
Datasets: bdcrops.csv (19 rows) + bd_district_crop.csv (505 rows)
6 queries
Key findings:
Mustard/Rapeseed leads oil potential at 1.08 million MT
Mung Bean and Mustard fit best into seasonal gaps
Bangladesh can generate 12.7 million MT of additional crops
and 2.0 million MT of edible oil from existing untapped land
Oilseed expansion concentrated in Khulna and Rajshahi regions

PART 4 — THE SOLUTION MODEL
3 conceptual design queries (no SQL)
Key findings:
Ward Hub: integrated facility within 1 km of every farmer
Contains warehouse, seed bank, training center, help desk,
input shop, processing unit, and buy-sell center
Union Hub: coordination and distribution layer
District Hub: strategic control and inter-regional food movement
Three tiers form a complete national food grid

RECAP

PART 5 — THE BUSINESS CASE
Datasets: hub_costs.csv (8 rows) + impact_projection.csv (11 rows)
+ megaprojects.csv (8 rows) | 12 queries

Key findings:
Total investment: approximately 20,615 crore BDT
Break-even: Year 7
Net benefit by Year 10: 16,793 crore BDT
ROI: 2.05x more than doubles every taka invested
Cost per beneficiary: only 944 BDT versus 10,000-70,000 BDT for major megaprojects
151,900 permanent jobs + 237,800 seasonal jobs created

PART 6 — THE PILOT PLAN
Dataset: pilot_districts.csv (5 rows) | 4 queries

Key findings:
Total pilot cost: approximately 1,200 crore BDT
5 districts representing 5 distinct agro-zones
Year 1 output: 46,500 MT additional oil production
10,800+ direct jobs in Year 1
155 crore BDT annual revenue from pilot alone
85% of investment at ward level directly farmer-facing

PART 7 — THE GLOBAL WARNING
Datasets: global_food_prices.csv (10 rows) + export_bans.csv (8 rows) + country_comparison.csv (10 rows) | 3 queries

Key findings:

Export bans trigger price increases of 28% to 180%
Indonesia's 2-month palm oil ban alone caused a 42% price surge

FAO vegetable oil index rose from 85 to 171 between 2015-2022
Price spikes align directly with COVID-19, Russia-Ukraine war,
and unilateral export bans

Bangladesh food security rank: 84 out of 113 countries
Thailand ranks 39, Vietnam ranks 55

Forex reserves cover only 2.2 months of food imports
India: 11.5 months | Thailand: 7 months

Bangladesh faces a dual structural risk:
Highest import dependency + Lowest buffer capacity among peers
This requires immediate systemic intervention

KEY NUMBERS: THE PROBLEM

IMPORT CRISIS

$3-4 billion	Annual food import cost
90%	Edible oil import dependency
28% to 180%	Price increase from export bans
84 out of 113	Bangladesh food security ranking
2.2 months	Forex reserve import coverage

LAND UNDERUSE

802,000 hectares	Rice-fallow land empty every year
64 districts	Mapped and analyzed
Mymensingh	Highest untapped land priority district

CROP GAP

12.7 million MT	Additional crop production possible
2.0 million MT	Additional oil from existing untapped land
1.08 million MT	Mustard/rapeseed oil potential alone
505	District-crop combinations analyzed

GLOBAL RISK

8 export bans	Directly impacted Bangladesh (2019-2024)
171	FAO vegetable oil index peak (2022)
11%	Bangladesh food inflation peak (2022)

Key Numbers: The Solution and Economics

INFRASTRUCTURE

33,000	Approximate ward hubs needed
1 km	Average farmer distance to hub
5 variants	Customized by region
7 sections	Inside every ward hub
3 tiers	Ward → Union → District

INVESTMENT

20,615 crore BDT	Total national investment
1,200 crore BDT	5-district pilot cost
944 BDT	Cost per beneficiary (lowest nationally)

RETURNS

2.05x	10-year return on investment
Year 7	Break-even point
16,793 crore BDT	Net benefit by Year 10

IMPACT

50%	Oil import dependency after 10 years
80%	Farmer income increase by Year 10
25%	Consumer food price decrease
151,900	Permanent jobs created
237,800	Seasonal jobs created
99%	Rice self-sufficiency by Year 10

THE PROJECT

11 datasets	788 total rows
42 queries	39 SQL + 3 conceptual
8 presentations	Parts 1 through 8
Tools	SQL and Power BI only

WHAT THIS PROJECT PROVED

ONE CORE FINDING ACROSS ALL 42 QUERIES:

Bangladesh's food insecurity is a system efficiency problem, not a production problem.

The land exists.
The crops exist.
The varieties exist.
The farmers exist.

What is missing is the infrastructure to connect them.

FOUR PROBLEMS PROVEN WITH DATA:

1 Import trap — billions spent on what can be grown at home
2 Land underuse — 802K hectares idle every year
3 Missing value chain — middlemen capture what farmers earn
4 Global vulnerability — highest dependency, lowest buffer

ONE SOLUTION PROVEN WITH DATA:

KrishiGrid delivers 2.05x ROI in 10 years.
Breaks even by Year 7.
Costs only 944 BDT per beneficiary.
Serves 170 million people.

AS A DATA ANALYST:

Data does not just describe problems.
Data can design solutions.

Every query in this project answered a decision:
Where is the risk? Where is the land? What to grow?
What does it cost? When does it pay back?
What happens if we do nothing?

That is what data analysis is for.

ALL DATA SOURCES

BANGLADESH GOVERNMENT
Bangladesh Bank — bb.org.bd
Import payments, forex reserves

BBS (Bangladesh Bureau of Statistics) — bbs.gov.bd
Statistical yearbook, agricultural census

DAE (Dept. of Agricultural Extension) — dae.gov.bd
Crop statistics, Krishi Diary

SRDI — srdi.gov.bd
Soil mapping, land classification

BARC — barc.gov.bd
Agro-ecological zones, land capability

RESEARCH INSTITUTIONS
BRRI — brri.gov.bd
Rice variety, yield, and duration data

BARI — bari.gov.bd
Oilseed and crop variety research

BINA — bina.gov.bd
Short-duration variety development

DATA RELIABILITY
All 11 datasets contain a data_reliability column.
Values: actual, estimated, interpolated.
All estimates cross-referenced for consistency.
This is a simulation, not an official policy document.

INTERNATIONAL
FAO FAOSTAT — fao.org/faostat
Trade database, food price index

World Bank — worldbank.org
Commodity prices, agricultural reports

USDA FAS — fas.usda.gov
Bangladesh commodity and trade data

The Economist GFSI — economist.com/gfsi
Global Food Security Index scores

IMF — imf.org
Forex reserve data, economic outlook

WFP — wfp.org
Global acute food insecurity data

THANK YOU

KRISHIGRID BANGLADESH

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"Data does not just describe problems.
Data can design solutions."

FULL REPOSITORY

github.com/rafijurrahman/krishigrid-bangladesh

All datasets, SQL files, Power BI dashboards, PDF exports, and LinkedIn post drafts.
Open source. Free to use with attribution.

SERIES SUMMARY

Part 1 The Import Crisis
Part 2 The Hidden Land Opportunity
Part 3 The Crop Revolution
Part 4 The Solution Model
Part 5 The Business Case
Part 6 The Pilot Plan
Part 7 The Global Warning
Part 8 The Complete Journey

CONNECT

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